CISS240: Introduction to Programming
Quiz q1903

Name: eagraham1@cougars.ccis.edu Score:

This is a closed-book, no compiler, 5 minute quiz.

Q1. Write a function `sign()` that accepts a `double x` and return 1, 0, or −1 (integer) depending on whether `x` is greater than 0, is equal to 0, or is less than 0 respectively. In other words, the following code segment

```
std::cout << sign(5.3) << ' ' << sign(0.0) << ' ' << sign(-1.1);
```

will produce the output

```
1 0 -1
```

Answer:

```
int sign(double x)
{
    if (x > 0)
        return 1;
    else if (x == 0)
        return 0;
    else
        return -1;
}
```

Q2. Write the function `swap` so that swaps the values of two `double` variables:

```
double x = 1.1, y = 2.2;
swap(x, y);     // x becomes 2.2 and y becomes 1.1
```

Answer:

```
void swap(double &x, double &y)
{
    double temp = x;
    x = y;
    y = temp;
}
```

Q3. Write the function `cut_tens` so that the following

```
int x = 1234, digit;
digit = cut_tens(x); // digit 3 is returned and x becomes 124
```

will "cut out" the digit in the tens place from `x` and return this digit.
Answer:

```
int cut_tens(int &x)
{
    int digit = (x / 10) % 10;
    x = (x / 100) * 10 + (x % 10);
    return digit;
}
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.